

Exercise 2 : SQL Aggregate Functions & Operators

Q1 SELECT DISTINCT department
FROM students;

department
IT
HR
Finance

2 SELECT department;
AVG (age) AS avg-age
FROM students
GROUP BY department;

department	avg-age
IT	20,5
HR	22
Finance	23

3 SELECT department,
COUNT (*) AS student-count
FROM students
GROUP BY department
HAVING COUNT (*) > 1;

department	student-count
IT	2
HR	2

4) SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5) SELECT student_id, name, age, department
FROM students
WHERE department = 'IT' OR 'HR'
AND age > 21;

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6 SELECT department,
SUM(credits) AS total-credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5

department	total-credits
IT	11

7 SELECT course_id, course_name, department, credits
FROM courses
WHERE credits NOT 4;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8 SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9 SELECT MAX (grade) AS max_grade,
MIN (grade) AS min_grade,
AVG (grade) AS avg_grade
FROM enrollments;

max_grade	min_grade	avg_grade
90	78	84.6

10) SELECT course_id ,
COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id ;

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11) SELECT department ,
SUM(salary) AS total_salary ,
SUM(Bonus) AS total_bonus
FROM salaries
GROUP BY department ;

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

12) SELECT department ,
AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55 000 ;

department	avg_salary
IT	61 000
Finance	70 000

FROM Salaries
 WHERE (salary + bonus) > 60 000 ;

employee-id	name	salary	bonus	total-compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5 500	67 500

14) SELECT department ,
 SUM (budget) AS total-budget ,
 AVG (budget) AS avg-budget
 FROM projects
 GROUP BY department
 HAVING AVG(budget) > 70 000 ;

department	total-budget	avg-budget
IT	270 000	135 000
Finance	80 000	80 000

15) SELECT project_id , project_name , department , budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
AND department NOT 'Marketing' ;

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000